

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<thought>

To solve this problem , we need to understand the role of CD 2 1 in the immune response and which virus ’s pathogenesis would be affected by B cells lacking CD 2 1 . CD 2 1 , also known as P N A , is a complement regulatory protein expressed on B cells and certain other cell types . It plays a crucial role in complement-dependent T-cell activation (C C T A) which is vital for the activation of B cells and production of antibodies against certain pathogens .

</thought>

< answer >

A . Epstein-B arr virus (EB V) </answer>

<thought>

Ep stein-B arr virus (EB V) interacts with CD 2 1 during its lifecycle to activate immune response . However , if B cells are modified to not express CD 2 1 , they will be unable to fully activate complement-dependent T cell activation , thus affecting the immune response to EB V antigens . The presence of EB V can lead to certain cancers and diseases like Infectious Mon on ucle osis (IM) due to the dys regulated immune response .

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